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Title:

- Photonic_Toroidal_Vortex_Architecture

Description:

- An_exact,_semi-classical,_geometric_alternative_to_probabilistic_theories._It_replaces_point_abstractions_with_a_local,_mechanical_conservation-of-action_principle,_within_a_structured_rotating-photon-framework.

Aim:

- Obtain_a_visualisable_mechanical_model_of_the_unobservable_world_beyond_the_senses
- Interpret_anthropocentric_theories_using_a_universal_model

Core_achievements:

- obtains_energy_levels_for_an_electron_without_an_external_potential
- reproduces_the_Sommerfeld-Dirac_fine_structure_hydrogen_atom_energy_levels_to_2_parts_in_10¹⁰
- obtains_the_hyperfine_half_splitting_for_hydrogen_across_30_states_with_a_median_error_of_0.0014_MHz
- provides_explicit_local_physical_mechanisms_for_intrinsic_spin,_the_Lorentz_force_deflection,_mass_generation,_charge_invariance,_and_Bose_Einstein_statistics_with_distinguishable_photons

Main_research_and_documentation:

- <https://barryispuzzled.com/physics>

Relevant_publications:

- Reinterpretation_of_the_Grangier_experiment_using_a_multiple-triggering_single_photon_model,_Modern_Physics_Letters_B,(2023),_DOI:_10.1142/S0217984923500422
- A_photonic_toroidal_vortex_model_of_the_hydrogen_atom_fine_structure,_Quantum_Studies:_Mathematics_and_Foundations,(2025),_DOI:_10.1007/s40509-025-00364-9

Preprints:

- Geometrical_interpretation_of_the_hydrogen_atom_hyperfine_structure,(2026),_DOI:_10.20944/preprints202603.1719.v1
- The_Lorentz_force_and_the_nature_of_charge_from_a_photonic_toroidal_vortex_model,(2026),_DOI:_10.20944/preprints202603.2233.v2

- A_heuristic_model_of_the_Bose-Einstein_distribution_with_distinguishable_photons,(2026),_DOI:_10.20944/preprints202605.1447.v1

Author:

Name: Dr Barry R. Clarke

Curriculum_vitae:

- The_Vortex_Atom,_World_Scientific_Publishing,(2021)
- The_Quantum_Puzzle,_World_Scientific_Publishing,(2017)
- Puzzle_compiler_for_the_Daily_Telegraph_with_over_1800_mathematics_and_logic_puzzles
- Challenging_Logic_Puzzles_Mensa,_Sterling,(2003),_over_100,000_sold
-

MSc_by_thesis_in_matrix_mechanics_perturbation_methods_with_two_publications_in_J._Phys._A

- PhD_in_Shakespeare_attribution_studies

website:

- <https://barryispuzzled.com>

HS:

type: fundamental_construction_unit

description: finite-length_string

Sp-1:

derived_from: HS

description: optical_spin_angular_momentum_(optical_SAM)

structure: HS_helical_trajectory

Sp-2:

derived_from: Sp-1

description: optical_orbital_angular_momentum_(optical_OAM)

structure: Sp-1_helical_trajectory

Sp-3:

derived_from: Sp-2

description: toroidal_rotation

structure: Sp-2_rotation_around_toroidal_axis

constants:

alpha: fine_structure_constant

c: speed_light

m₀: electron_rest_mass

m'₀: proton_rest_mass

h: Planck's_constant

ħ: h/(2*π)

M: m₀/m'₀

Ry : Rydberg_frequency
An: nuclear_mass_number
Zn: nuclear_charge

variables:

na: Sommerfeld_azimuthal_quantum_number
nr: Sommerfeld_radial_quantum_number
n: total quantum number
- $n = n_a + n_r$
r2o: Sp-2_radius
- $r_{2o} = 3.86159268 \cdot 10^{-13} \text{ m}$
r3o: Sp-3_radius
- $r_{3o} = 5.291631210153 \cdot 10^{-11} \text{ m}$
v2: Sp-2_speed
v2f: Sp-2_field_speed
v3: Sp-3_speed
v3f: Sp-3_field_speed
p2f: Sp-2_field_momentum
p3f: Sp-3_field_momentum
xbar: bound-state_distance_between_proton-electron_toroidal_centers_as_fraction_of_r3o
dbar2: bound-state_distance_between_proton-electron_Sp-2_centers_as_fraction_of_r3o
E3: fine_structure_formula_with_reduced_mass
E3D: fine_structure_formula_with_reduced_mass_and_velocity_adjustment
nu_midpoint: hyperfine_midpoint_frequency_(MHz)
nu_magnetic_potential: magnetic_potential_frequency_(MHz)
nu_shift: hyperfine_half-interval_shift_from_midpoint_(MHz)
nu: frequency_(MHz)_includes_fine_structure_with_reduced_mass_and_velocity_adjustment,_magnetic_potential,_and_hydrofine_half-interval_shift
capital_gamma: searchable_parameter_in_fine_structure_velocity_adjustment
A: Searchable_parameter_in_magnetic_potential
B: Searchable_paremeter_in_magnetic_potential
k: muliplier_in_hydrofine_half-shift_(k=1_for_nS1/2_states)

Fundamental_postulates:

- Sp-1_and_Sp-2_rotations_tend_to_preserve_action_h
- rotational_and_translational_action_are_interchangeable_when_Sp-1_or_Sp-2_action_is_overloaded_or_underloaded
used_in:
-
dispersive_media_when_photon_speed_slows_due_to_translational_action_redistributed_into_Sp-1_action
-
hydrofine_shift_when_changing_magnetic_momentum_field_overloads_or_underloads_electron_Sp-2_action_into_or_out_of_Sp-3_action

- Lorentz_force_when_Sp-2_circuit_in_toroid_is_underloaded_or_overloaded_due_to_field_gradient_across_it_and_change_in_Sp-3_translational_action_gives_deflection

Attraction_mechanism:

- proton_and_electron_Sp-3_have_opposite_rotations
- proton_Sp-3_field_momentum_intrudes_in_electron_Sp-3_rotation
- electron_Sp-3_action_reduced_by_field_momentum_so_toroid_attempts_to_find_next_lowest_integer_action
- electron_seeks_stronger_proton_field_to_reduce_action_further
- electron_Sp-3_energy_redistributed_into_translational_motion_towards_proton
- translational_energy_emitted_as_radiation_at_bound-state_distance

Repulsion_mechanism:

- same_charges_have_same_sense_Sp-3_rotations
- action_in_target_Sp-3_is_overloaded_by_intruding_Sp-3_field_momentum
- toroid_attempts_to_find_next_lowest_integer_action
- toroid_seeks_weaker_source_field_to_reduce_action_further
- Sp-3_energy_redistributed_into_translational_motion_away_from_source

rest_mass:

derived_from: Sp-2

structure: circular_Sp-2_rotation

condition: inversely_proportional_to_Sp-2_radius

equation: $m_0 = h/(2\pi r_2 \omega c)$

relativistic_mass:

derived_from:

- Sp-2
- Sp-3

condition: Sp-2_helical_trajectory_due_to_Sp-3_momentum_component

magnetic_momentum_field:

derived_from: Sp-2

structure:

- permanent_field_emission_due_to_Sp-2_rotation
- p_{2f} _field_momentum
- r_{2f} _field_radius
- radius_taken_from_Sp-2_center
- mass_of_field_invariant
- speed_inversely_proportional_to_ r_{2f}
- field_cuts_line_joining_source-target_Sp-2_centers_at_ $\pi/4$
- field_lines_are_parallel

used_in:

- changing_p2f_field_momentum_creates_hyperfine_splitting

formula:

- $p_{2f} = \hbar/r_{2f}$

- $v_2 = c \cdot \sqrt{1 - \alpha}$

electric_momentum_field:

derived_from: Sp-3

structure:

- permanent_field_emission_due_to_Sp-3_rotation

- $p_{3f_field_momentum}$

- $r_{3f_field_radius}$

- radius_taken_from_Sp-3_center

- electric_field_momentum_perpendicular_to_magnetic_field_momentum

used_in:

- Coulomb_field

- Ampere's law: aligned_Sp-3_field_rotations_around_current-carrying_conductor

formula:

- $p_{3f} = \hbar \cdot \alpha / (r_{2f} \cdot \sqrt{1 - \alpha^2})$

- $v_3 = \alpha \cdot c$

dbar2:

name:

- bound-state_distance

derived_from:

- electron_Sp-3_residual_energy_after_field_intrusion

- proton_Sp-3_momentum_field_energy

condition: electron_Sp-3_residual_energy == proton_Sp-3_momentum_field_energy

used_in:

- fine_structure_velocity_adjustment_D-function

- magnetic_potential

- hyperfine_shift

- neutron_construction

formula:

- $\bar{d} = \sqrt{x^2 + (1 - M)^2}$

charge:

derived_from:

- Sp-2

- Sp-3

- Lorentz_force_deflection_analysis

structure:

- view_rotations_in_v2_and_v3_directions

- two_chiralities_for_each_charge

- negative_charge: clockwise_Sp-2_and_counter_clockwise_Sp-3

- negative_charge: counter_clockwise_Sp-2_and_clockwise_Sp-3

- positive_charge: counter_clockwise_Sp-2_and_counter_clockwise_Sp-3
- positive_charge: clockwise_Sp-2_and_clockwise_Sp-3

principle:

- charge_invariance_when_viewing_it_moving_away_from_other_side

fine_structure_reduced_mass:

formula:

- MHz_frequency
- $Y = nr^2 + \sqrt{na^2 - \alpha^2}$
- $\mu = m_0/(1 + M)$
- $E3 = (10^{-6}/h) * (\mu / \sqrt{1 - \alpha^2/(2*(Y^2 + \alpha^2))}) * \alpha^2 * c^2 / (2*(Y^2 + \alpha^2))$

fine_structure_velocity_adjustment:

formula:

- MHz_frequency
- $D = (1 + \text{capital_gamma} * M/n)^2$
- $YD = nr^2 + \sqrt{na^2 - D * \alpha^2}$
- $\mu = m_0/(1 + M)$
- $E3D = (10^{-6}/h) * (\mu / \sqrt{1 - D * \alpha^2/(2*(Y^2 + D * \alpha^2))}) * D * \alpha^2 * c^2 / (2*(Y^2 + \alpha^2))$
- α^2 _in_E3_replaced_by_ $\alpha^2 * D$
- relative_speed_adjustment_of-electron_along_line_joining_Sp-2_centers

magnetic_potential:

formula:

- MHz_frequency
- $\nu_{\text{magnetic_potential}} = A * (\text{dbar}^2)^B$

nature:

- with_B=-3.000_(to_three_decimal_places_for_all_states)_it_resembles_a_dipole_moment
- boundary_term_that_remains_when_differentiated_field_momentum_with_respect_to_spatial_coordinates_is_integrated_over_trajectory_from_infinity_to_bound-state_distance_dbar2
- cumulative_effect_of_Sp-2_field

hyperfine_midpoint:

- used_instead_of_the_hyperfine_centroid

used_in: base-line_for_equal_magnitude_hyperfine_half-interval_shift_each_side

formula:

- MHz_frequency
- $\nu_{\text{midpoint}} = E3D + \nu_{\text{magnetic_potential}}$

parameter_constraints:

B:

- capital_gamma_in_E3D_is_varied
- found_by_minimizing_difference_between_experimental_value_of_nu_midpoint_and_E3D

- for_all_60_hydrogen_states
- B = -3.000 (3_decimal_places)

A:

-

once_B_is_found_A_is_lined_up_on_lowest_n_of_set_for_example_1S1/2_in_nS1/2,_it_must_then_work_for_all_other_states_in_that_set

- it_then_has_to_work_on_other_five_tested_states_in_that_set

-

two_values_for_each_states_with_A>0_and_A<0,_corresponding_to_two_values_of_capital_gamm
a_<0_and_>0

universal_scaling_law:

- A/capital_gamma = -28651713/(An*Zn)
- law_has_0.03%_error_for_deuterium
- law_has_0.04%_error_for_tritium
- law_has_0.05%_error_for_3He+
- law_has_0.01%_error_for_7Li2+

hyperfine_shift:

- two_cases_per_stateDepending_on_whether_the_proton-electron_have_same_or_opposite_sense_rotations
- same_sense_proton-electron_Sp-2_rotations_as_they_approach_gives_red-shift,_capital_gamma_>0
- opposite_sense_proton-electron_Sp-2_rotations_as_they_approach_gives_blue-shift,_capital_gamma_<0

formula:

- nu_shift_value_can_be_plus_or_minus
- $$\text{nu_shift} = 2 * k * \pi * n^2 * \alpha^3 * R_y / (\text{sqr}(2) * Y^2 * \text{sqr}(1 + \alpha^2 / Y^2)) * (1 - \alpha^2)^{(3/2)} * \text{dbar}^2^3$$
- k=1_for_nS1/2_states_but_varies_for_other_sets_due_to_Sp-2_eccentricity_which_reduces_field_receptivity

nature:

- effect_of_changing_Sp-2_field_at_electron_Sp-2_location_in_one_electron_Sp-2_time_period

complete_calculation:

formula:

- $$\text{nu} = E_{3D} + \text{nu_magnetic_potential} + \text{nu_shift}$$